TEXORA · LECTURE SHEET

Modern Fibre Testing Equipment: AFIS and HVI

Two instrument families replaced a bench of slow classical tests: HVI measures a beard and a bundle at production speed for bale classification, while AFIS takes the sample apart fibre by fibre to expose neps, trash and the full length distribution. Knowing which instrument answers which question is the skill.

TE-3103 · QUALITY EVALUATION OF FIBRE AND YARN · TOPIC 9 · LEVEL 3 TERM 1

1. Why instruments changed

The classical methods of Topics 3–8 are accurate but slow: a sorter length analysis or a swelling-test maturity count occupies a skilled operator for the better part of an hour on one sample. Cotton trading and modern spinning need a full fibre profile on *every bale* and process checks on *every shift*. The answer was automation in two complementary directions:

- **High Volume Instrument (HVI)** lines — beard/bundle measurements, a complete classification in under a minute per sample, built for bale-by-bale testing.
- **AFIS (Advanced Fibre Information System)** — individualisation of the sample and fibre-by-fibre electro-optical measurement, slower (minutes per sample) but revealing distributions and particle counts no bundle test can see.

2. HVI — modules and outputs

A conditioned sample (about 150–200 g drawn from the bale, or a sliver sample) is presented to a line of modules, each applying one classical principle at speed:

MODULE	PRINCIPLE (CLASSICAL ANCESTOR)	OUTPUTS
Length / strength	Fibrograph beard scan; 3.2 mm-gauge bundle break (Stelometer principle)	UHML (mm or in), mean length, uniformity index UI %, bundle strength (g/tex), elongation %
Micronaire	Airflow through a compressed plug	Micronaire units (Topic 5 — combined fineness/maturity index)

Colour	Reflectance measurement against a white standard	Rd (reflectance, brightness) and +b (yellowness) → colour grade
Trash	Camera scan of the specimen surface	Trash particle count, % surface area covered → leaf grade estimate
Moisture (where fitted)	Electrical / NIR sensing of the sample	Sample moisture % — results are corrected / flagged to the conditioned state

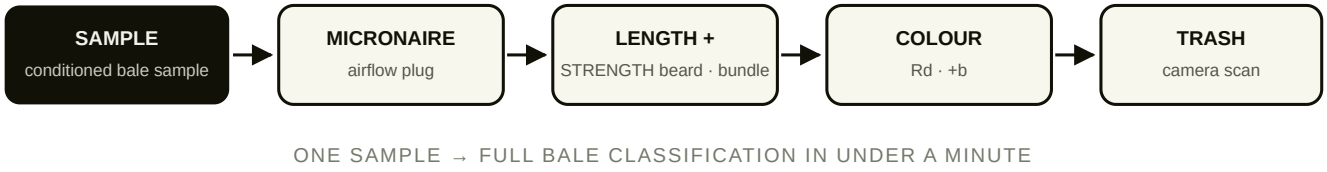


Figure 1 — HVI module flow. Each module is a classical method mechanised; the line's value is that every bale receives every test.

WORKED EXAMPLE — READING AN HVI REPORT

Bale report: UHML 28.2 mm; UI 82.0 %; strength 30.5 g/tex; elongation 6.8 %; micronaire 4.2; Rd 76; +b 8.5; trash area 0.6 %.

Mean length = UI × UHML / 100 = 0.82 × 28.2 = **23.1 mm**.

Reading: a medium-long upland cotton (28.2 mm ≈ 1.11 in), uniform (UI 82 % is a good commercial figure), strong (30.5 g/tex bundle tenacity), micronaire inside the preferred 3.5–4.9 base range, bright and fairly clean. This bale supports medium-fine counts; nothing in the profile warns the spinner. Contrast in the Practice section.

Calibration is the HVI contract. Lines are calibrated with **calibration cottons** of certified values (length/strength cottons and micronaire cottons) at the start of work and checked through the day; inter-laboratory round trials keep different lines in agreement. An uncalibrated HVI produces precise nonsense at high speed.

3. AFIS — fibre-by-fibre information

AFIS takes a small specimen (about 0.5 g), opens it with a miniature carding/individualising unit into an air stream, and carries the separated entities — single fibres, neps, trash and dust particles — past electro-optical sensors that size and classify each one. From thousands of individual events (≈ 3,000–5,000 fibres per replicate) it builds:

- **Length:** mean length by number L(n) and by weight L(w), their CVs, and short fibre content by number SFC(n) and by weight SFC(w) — the full distribution of Topic 3.
- **Neps:** total nep count per gram and mean nep size, with seed-coat neps separated (Topic 6).
- **Trash:** trash and dust particle counts per gram by size class, and a derived visible foreign matter estimate (Topic 7).
- **Fineness and maturity** (with the appropriate module): fineness in mtex and a maturity index (Topic 5).

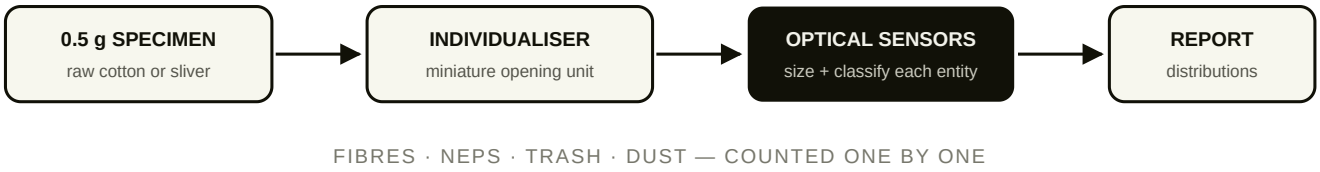


Figure 2 — AFIS flow. Individualisation is the whole secret: once entities travel singly, sensors can count and size what bundle tests can only average.

WORKED EXAMPLE — USING AFIS TO JUDGE A CARD

AFIS on card feed: neps 285/g, SFC(w) 9.5 %. AFIS on the card sliver from the same material: neps 48/g, SFC(w) 10.8 %.
Nep removal efficiency = (285 − 48)/285 × 100 = **83.2 %** — excellent carding.
But SFC(w) *rose* by 1.3 percentage points: the card is also breaking fibres. The balanced verdict — cleaning well, treating fibre roughly — points the mechanic to clothing condition and settings that open harshly, a diagnosis no single HVI bale figure could produce. AFIS earns its place as the process-diagnosis instrument.

4. HVI or AFIS — choosing correctly

QUESTION	INSTRUMENT	WHY
Classify every bale for purchase / laydown	HVI	Speed; strength and micronaire included; calibrated trade language
Why is this sliver neppy / this yarn specky?	AFIS	Nep counts + seed-coat split + SFC expose the cause
Bundle strength for a specification	HVI	AFIS does not measure strength

Full length distribution and short fibre content	AFIS	HVI gives UHML/UI summaries of the beard only
Maturity separate from fineness	AFIS (or double-compression airflow)	HVI micronaire deliberately combines them

5. Common faults in instrumental testing

- Testing unconditioned samples — HVI length/strength and micronaire all shift with moisture; samples are conditioned (Topic 1) and laboratory atmosphere is controlled.
- Skipping calibration checks because "the line was fine yesterday".
- Sampling one spot of the bale (Topic 1): the instrument is fast, the sample is still the weak link.
- Comparing numbers across instrument families (HVI UHML vs AFIS L(w)) as if they were the same statistic — they are cousins, not twins; each mill should track its own conversion on its own cottons.

Glossary

HVI — High Volume Instrument line for bale classification. **AFIS** — Advanced Fibre Information System; individual-fibre measurement. **UHML** — upper-half mean length. **UI** — uniformity index = mean length / UHML × 100. **Rd / +b** — HVI colour coordinates: reflectance and yellowness. **Calibration cotton** — reference cotton of certified property values. **SFC(n), SFC(w)** — short fibre content by number / by weight. **Individualisation** — separation of a sample into single entities for counting.

Key points

• HVI = speed and trade classification: length (UHML/UI), strength, micronaire, colour, trash on every bale.
• AFIS = diagnosis: distributions, SFC, neps (incl. seed-coat), trash/dust counts, fineness and maturity.
• Both stand on classical principles; both stand or fall by sampling, conditioning and calibration.
• Never substitute one family's statistic for another's without a locally established relation.

Practice

1. An HVI report gives UHML 29.5 mm and UI 83 %. Find the mean length.
2. A bale shows micronaire 3.3, strength 27 g/tex, UHML 27 mm. A second bale shows micronaire 4.6, strength 31 g/tex, UHML 29 mm. Which would you choose for a fine dyed shirting, and what extra

test decides whether the first bale's low micronaire is a virtue or a warning?

3. Card feed AFIS neps 320/g; sliver 96/g. Find the nep removal efficiency and comment.
4. Why can AFIS not replace HVI in bale trading, despite giving more information per sample?

Answer key

1. Mean length = 0.83 × 29.5 = **24.5 mm**.
2. On HVI alone, the second bale: longer, stronger, micronaire in the preferred range. The first bale's 3.3 may mean fine-and-mature (excellent for shirting) or coarse-and-immature (a nep and white-speck risk). The deciding test is a maturity measurement separated from fineness — AFIS maturity index (or double-compression airflow / swelling test).
3. Efficiency = (320 − 96)/320 × 100 = **70 %** — acceptable but not good; check clothing condition, settings and feed rate before blaming the cotton.
4. Trading needs a complete, standardised profile — including bundle strength, which AFIS does not measure — on every bale within seconds-to-a-minute, in a calibration language all parties share. AFIS takes minutes per small specimen and answers diagnostic questions; the two are complements, not rivals.
